

CPG-Focused Datasets for 3 Models

Outside the List

1. [Instacart Market Basket Analysis](#): 3M+ grocery orders from 200K+ users, with **reorder flags**, days-since-prior-order, product department, and aisle hierarchy: dairy, snacks, personal care. These are all CPG categories. It can support:

- **RFM**: Recency (days_since_prior_order), Frequency (order count per user), Monetary (basket size × product)
- **Launch Planner**: Product-level demand patterns, reorder rates by category for forecasting new similar products
- **Budget Allocation**: Promotion flag on orders allows measuring promoted vs. organic conversion

2. [Dunnhumby — The Complete Journey](#): 2 years of household-level grocery transactions from 2,500 frequent shoppers at a CPG retailer. Includes **campaign data, coupon redemptions, demographics, and causal data** (weekly mailer features, in-store displays). Also supports:

- **RFM**: Full transaction history with household IDs, spend, frequency
- **Budget Allocation**: Campaign × coupon redemption data allows model marketing ROI and channel effectiveness directly
- **Launch Planner**: Causal data (promotions, displays) tied to sales lift. This is ideal for simulating launch campaign timing

3. [Store Sales — Time Series Forecasting \(Corporación Favorita\)](#): 125M+ observations from a large grocery chain, with product categories: Grocery, Beverages, Dairy, Personal Care. It also has store metadata, promotions, holidays, and oil price data. **Suitable for Launch Planner Modeling**:

- **Launch Planner**: Multi-store, multi-product daily sales with promotion effects and seasonality, which is exactly what we need for demand forecasting and inventory timing
- **RFM**: Aggregatable to store/product family level for segmentation simulation
- **Budget Allocation**: Promotion flag per item enables measuring promotional ROI by product family

4. [Crisp CPG Sample Data \(BigQuery\)](#) Supplementary dataset: Purpose-built CPG retail sales and supply chain data (US + EU) from Crisp, mimicking real brand-retailer data at SKU level. Free via Google BigQuery. **Good for validating Launch Planner inventory models** against realistic CPG supply chain patterns.

5. [FMCG Sales Demand Forecasting and Optimization](#) Supplementary dataset: Fast Moving Consumer Goods daily sales data designed for time-series forecasting and decision analysis. Useful as a secondary dataset for **Launch Planner demand modeling**.

Inside the List

6. [UCI Online Retail Dataset](#) Supplementary dataset: While not CPG-labeled, it's a UK gift/homeware retailer with high frequency consumable repurchase patterns. It is solid for RFM prototyping.

Dataset	Link	Primary Use	Secondary Use	CPG Fit
Instacart Market Basket	Link	RFM (reorder behavior)	Launch Planner (category demand)	Pure CPG grocery
Dunnhumby Complete Journey	Link	Budget Allocation (campaign ROI)	RFM (household segments)	CPG + marketing campaigns
Corporación Favorita (Store Sales)	Link	Launch Planner (demand forecasting)	Budget Allocation (promo lift)	Grocery/CPG multi-store
Crisp CPG Sample Data	Link	Launch Planner (supply chain)	Inventory modeling	Built for CPG brands
UCI Online Retail	Link	RFM baseline prototyping	Revenue simulation	General retail
FMCG Sales Forecasting	Link	Launch Planner (supplementary)	Demand optimization	FMCG is similar to CPG